Supplementary Information: Unveiling Mycobacterium marinum Cellular and Metabolic Remodeling Under Simulated Microgravity

Richard Barker[1,*], Lynn Harrison[2], and Joseph L. Clary[2]

[1]Department of Agricultural and Biological Engineering, Purdue University, West Lafayette, IN 47907, USA
[2]Department of Molecular and Cellular Physiology, LSU Health Shreveport, Shreveport, LA 71103, USA

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1 Supplementary Tables Overview

All empirical data tables are archived in the repository under `results_tables/` and available via Zenodo:

- **Table S1** (`Table_S1_raw_counts_matrix.tsv`): Complete raw transcript count matrix ($5,510$ genes$\times$ 9 samples) generated by `kallisto` v0.52.0 pseudoalignment of NextSeq 550 paired-end reads.
- **Table S2** (`Table_S2_normalized_log2cpm_expression.tsv`): Normalized $\log_2(\mathrm{CPM}+1)$ matrix across $4,964$ expressed genes.
- **Table S3** (`Table_S3_sample_metadata.tsv`): Experimental metadata for NASA OSDR OSD-528.
- **Table S4** (`Table_S4_differential_expression_3dclinostat_vs_1g.tsv`): 351 significant DEGs in 3D Clinostat vs Static 1g.
- **Table S5** (`Table_S5_differential_expression_rpm2_vs_1g.tsv`): 738 significant DEGs in RPM 2.0 vs Static 1g.
- **Table S6** (`Table_S6_differential_expression_3dclinostat_vs_rpm2.tsv`): 242 DEGs comparing simulation kinematics.
- **Table S7** (`Table_S7_wgcna_module_assignments_goslim.tsv`): WGCNA module assignments, standardized GOSlim descriptors, and k_{within} hub scores.
- **Table S8** (`Table_S8_wgcna_module_eigengenes.tsv`): Module Eigengene profiles across all 9 samples.
- **Table S9** (`Table_S9_wgcna_module_trait_correlations.tsv`): Pearson correlation coefficients and asymptotic p-values.

- **Table S10** (`Table_S10_goslim_pathway_enrichment.tsv`): GOSlim pathway over-representation metrics (k/n overlaps and Benjamini-Hochberg FDR).
- **Table S11** (`Table_S11_full_gene_ontology_enrichment.tsv`): Comprehensive GO enrichment table across Biological Process and Molecular Function.
- **Table S12** (`Table_S12_tabpfn_loocv_predictions.tsv`): TabPFN leave-one-out cross-validation predictions and calibrated posterior probabilities.
- **Table S13** (`Table_S13_tabpfn_feature_importance.tsv`): TabPFN permutation feature importance rankings.
- **Table S14** (`Table_S14_cellular_metabolic_model_reactions.tsv`): Reaction-level GPR mapping across 32 enzymatic reactions.
- **Table S15** (`Table_S15_metabolic_subsystem_perturbation_index.tsv`): Quantitative Subsystem Perturbation Index (SPI) rankings.
- **Table S16** (`Table_S16_pan_microbial_osdr_compendium_78studies.tsv`): Compendium of 78 microbial spaceflight datasets in OSDR.
- **Table S17** (`Table_S17_cellular_metabolic_model_sbml.json`): SBML-compatible computational model schema.

2 Supplementary Figures Overview

All supplementary figures are archived as vector PDFs and high-resolution 300 DPI PNGs in `supplementary_figures/`:

- **Figure S1**: Systems architecture, biological model, simulation hardware, and analytical framework.
- **Figure S2**: Transcriptomic PCA biplot (70.1% PC1 variance) and volcano plot of 351 DEGs.
- **Figure S3**: Gene count distribution across 5 GOSlim co-expression modules and module-trait heatmap.
- **Figure S4**: TabPFN tabular foundation model benchmark under LOOCV, confusion matrix, and feature importances.
- **Figure S5**: Enriched GOSlim functional pathways with calibrated FAIR Blue-White-Red color fill.
- **Figure S6**: Intramodular hub connectivity (k_{within} vs k_{total}) and inter-module regulatory interactome.
- **Figure S7**: Pan-microbial spaceflight meta-analysis landscape across 78 OSDR studies and concordance matrix.
- **Figure S8**: Hexagonal trajectory radar mapping phenotypic concordance across 6 physiological axes.
- **Figure S9**: Multi-compartment cellular cross-section and Subsystem Perturbation Index (SPI) ranking.

3 Read Quality Control & Pseudoalignment Convergence

Pseudoalignment against *M. marinum* M strain (NC_010612.1) converged with high pseudoalignment efficiency across all 9 biological replicates (mean 1.21 million reads mapped per sample). Normalized counts were converted to log-CPM values using edgeR TMM library scaling. Zero synthetic data were utilized in any calculation.